

LifeBox NextGen

Employee Portal Development Responsibility Document

Version 1.0

1. Purpose of This Document

This document defines the responsibilities of the **Frontend Development Team** and the **Backend Development Team** while building the Employee Portal of the LifeBox NextGen system.

The goal of this document is to ensure clear separation of development responsibilities, avoid duplication of work, and ensure both teams collaborate effectively through defined APIs.

Frontend developers are responsible for building the user interface and user experience.

Backend developers are responsible for implementing system logic, database operations, security mechanisms, and APIs that power the frontend interface.

2. System Architecture Overview

The LifeBox NextGen Employee Portal consists of the following layers:

- Desktop Application Layer
- Frontend Application Layer
- Backend API Layer
- Database Layer

The frontend application communicates with the backend server using **secure REST APIs**.

All business logic and data operations are handled by the backend.

The frontend only consumes APIs and displays data.

3. Authentication Module

Backend Responsibilities

The backend team must implement the authentication system responsible for validating employee credentials.

The backend must provide APIs that allow users to log in, log out, and maintain secure sessions using token-based authentication.

Required APIs

Login API

POST /api/auth/login

Logout API

POST /api/auth/logout

Get Current User API

GET /api/auth/me

The backend must also implement password encryption and secure token generation.

Example scenario:

When a user submits login credentials, the backend validates them and returns a JWT token that will be used in future API requests.

Frontend Responsibilities

The frontend team must build the login interface.

This includes input fields for email and password, login submission logic, and handling authentication errors.

The frontend must store the authentication token received from the backend and attach it to future API requests.

Example scenario:

If login fails due to incorrect credentials, the frontend must display an appropriate error message.

4. Dashboard Module

Backend Responsibilities

The backend must provide APIs that supply dashboard information including employee profile data, upcoming events, and task summaries.

Required APIs

Get Employee Profile

GET /api/employees/profile

Get Upcoming Events

GET /api/events/upcoming

Get Team Online Status

GET /api/teams/online

Get Task Summary

GET /api/tasks/summary

These APIs provide the data required to populate the dashboard widgets.

Frontend Responsibilities

The frontend team must design the dashboard layout and display data retrieved from the backend APIs.

The dashboard must include widgets showing employee identity, upcoming events, team presence status, and task statistics.

Frontend developers must also ensure the dashboard updates dynamically when new data is received.

5. Tasks Module

Backend Responsibilities

The backend must implement the task management system and expose APIs for retrieving and updating tasks.

Required APIs

Get Employee Tasks

GET /api/tasks/my

Update Task Status

PUT /api/tasks/{task_id}

Upload Task Files

POST /api/tasks/{task_id}/files

These APIs allow employees to view tasks, update progress, and submit work-related files.

Frontend Responsibilities

The frontend team must create the task interface where employees can view assigned tasks, update status, and upload files.

Frontend must call backend APIs to retrieve tasks and send updates when employees change task status.

6. Attendance Module

Backend Responsibilities

The backend must implement attendance tracking and store employee check-in and check-out records.

Required APIs

Check In

POST /api/attendance/checkin

Check Out

POST /api/attendance/checkout

Attendance History

GET /api/attendance/history

Frontend Responsibilities

The frontend team must build the attendance interface displaying check-in time, working hours, and historical attendance records.

Frontend must trigger the check-in API when employees log in and display the attendance data returned by the backend.

7. Calendar Module

Backend Responsibilities

The backend must provide APIs for retrieving events and schedules.

Required APIs

Get Calendar Events

GET /api/calendar/events

Frontend Responsibilities

Frontend developers must implement a calendar interface displaying events, meetings, and deadlines retrieved from backend APIs.

8. Performance Module

Backend Responsibilities

The backend must compute productivity metrics based on monitoring data and task completion records.

Required APIs

Get Employee Performance

GET /api/performance/me

Frontend Responsibilities

The frontend must display performance data using charts or graphs showing employee productivity and work statistics.

9. Salary and Payslip Module

Backend Responsibilities

The backend must manage salary records and provide APIs for retrieving salary information.

Required APIs

Get Salary History

GET /api/salary/history

Download Payslip

GET /api/salary/payslip/{month}

Frontend Responsibilities

The frontend must display salary records and allow employees to download payslips provided by the backend.

10. Document Management Module

Backend Responsibilities

The backend must implement document storage and retrieval.

Required APIs

Get Documents

GET /api/documents

Upload Document

POST /api/documents/upload

Frontend Responsibilities

The frontend must allow employees to view available documents and upload required files.

11. Activity Tracker Module

Backend Responsibilities

The backend must store monitoring data received from the desktop application.

Required APIs

Submit Activity Log

POST /api/activity/log

Get Activity Logs

GET /api/activity/me

Frontend Responsibilities

The frontend must display activity logs in visual format showing application usage and idle time.

12. Messaging and Chat Module

Backend Responsibilities

The backend must implement real-time messaging services.

Required APIs

Send Message

POST /api/chat/send

Get Messages

GET /api/chat/messages

Frontend Responsibilities

The frontend team must build chat interfaces for individual and team communication.

13. Leave Management Module

Backend Responsibilities

The backend must handle leave requests and approval workflows.

Required APIs

Apply Leave

POST /api/leaves

Leave History

GET /api/leaves/history

Frontend Responsibilities

The frontend must provide leave request forms and leave history interfaces.

14. Helpdesk Support Module

Backend Responsibilities

The backend must implement a ticketing system to manage employee support requests.

Required APIs

Create Support Ticket

POST /api/support/tickets

Get Tickets

GET /api/support/tickets

Frontend Responsibilities

The frontend must provide forms allowing employees to submit support tickets and track their status.

Final Summary

Frontend Team Responsibilities

- Design and build user interfaces
- Display data returned from APIs
- Handle user interactions
- Send requests to backend APIs

Backend Team Responsibilities

- Develop business logic
- Implement database operations
- Provide secure APIs
- Manage authentication and data processing

Both teams must collaborate through the defined API contracts to ensure seamless integration of the Employee Portal system.